Data Duplication – Same data is used by multiple resources for processing, thus created multiple copies of same data wasting the spaces.

High Maintenance – Access control and verifying data consistency needs high maintenance cost.

Security – less security provided to the data.

So database systems became popular to overcome the above limitations of file system.

3.1.3 DBMS Concepts

There exist few standards that are applicable to all forms of database management Systems like Relational Database Management System (RDBMS) and Object Database Management System (ODBMS). All DBMS adheres to the following two basic concepts.

ACID Properties – The acronym stands for Atomicity, Consistency, Isolation and Durability. **A**tomicity follows the thumb rule "All or Nothing", while updating the data in database for the user performing the update operation. This update operation is called as transaction and it either commits (successful updating) or aborts (updating failure). **C**onsistency ensures that the changes in data value to be constant at any given instance. This property helps in the successful transaction. **I**solation property is needed during concurrent transaction. When multiple users do the transactions by accessing same object at the same time, the transaction is known as concurrent transaction. To prevent the conflict in database update, the transactions are isolated from other user and serialized. This is also known as Degree of Consistency. **D**urability is defined as the system's ability to recover all committed transactions during the failure of storage or the system.

Concurrency Control and Locking – It is the DBMSs mechanism used for data sharing. When the same data is shared among multiple users, proper access control is needed and privilege of changing the applications data item is controlled through Locking.

3.2 DBMS Database Models

The database technology came into existence in terms of models with relational and object-relational behavior. The major database models are listed below:

3.2.1 Hierarchical Database Model

The famous Hierarchical database model was IMS (Information Management

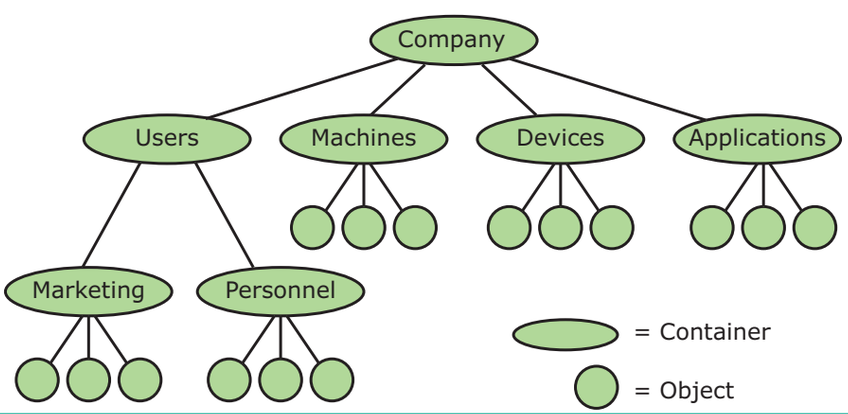


Figure: 3.1 Hierarchical database model

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